

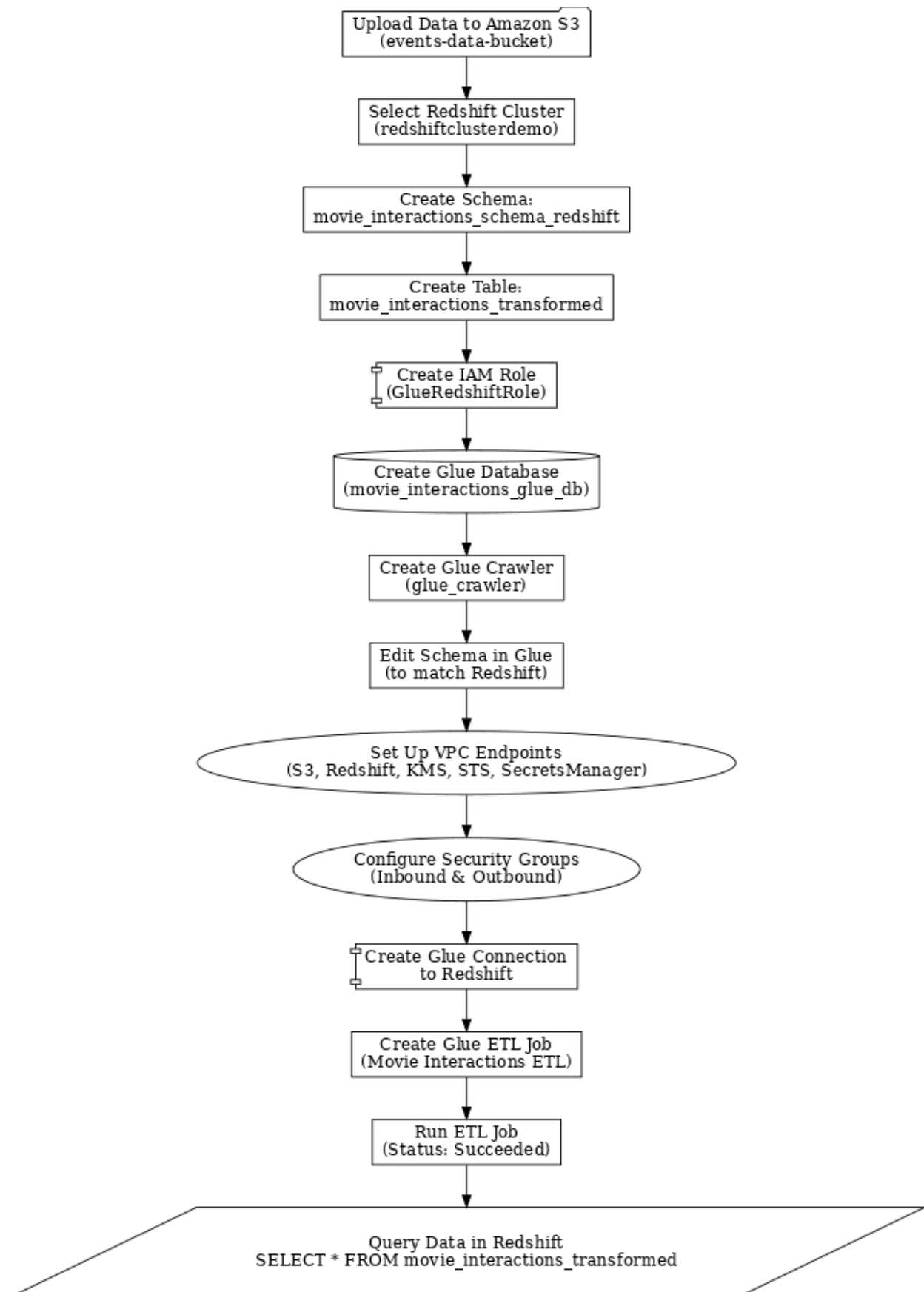


table).

So the flow is:

” Ask ChatGPT

Copy code

```
S3 (CSV) --[Crawler→Data Catalog]--> Glue ETL (transform) --[Glue connection]--> Redshift table
```

You then query the cleaned table from Redshift.

REDSHIFT:

Amazon Redshift > Clusters > Edit admin credentials

Edit admin credentials

Change admin user password

Admin password
Select an option to manage your admin password.

☐ Manage admin credentials in AWS Secrets Manager [Info](#)

☒ **Generate a password**
Amazon Redshift generates an admin password.

New admin user password

XXXXXXXXXX

☐ Show password

☐ Manually add the admin password
Manually enter the admin password.

Cancel **Save changes**

Save the password and we have to use in glue

CRAWLER:

The screenshot shows the AWS Glue console interface. On the left is a navigation pane with various AWS Glue services. The main content area displays the configuration for a crawler named 'glue_crawler'. The crawler is in a 'READY' state. The configuration includes the following properties:

Crawler properties			
Name	glue_crawler	IAM role	Gluedshiftrole
Description	-	Security configuration	-
Maximum table threshold	-	Database	movie_interactions_glue_db
		Lake Formation configuration	-
		Table prefix	raw_

Below the properties, there is a section for 'Data sources (1)'. It shows a table with the following columns: Type, Data source, and Parameters. The table contains one entry, which is partially visible.

The screenshot shows the 'Crawler runs' section for the 'glue_crawler'. It displays a table with the following columns: Name, Status, and Last updated (UTC). The table contains one entry, which is partially visible. The crawler is in a 'READY' state. The configuration includes the following properties:

Crawler properties			
Name	glue_crawler	IAM role	Gluedshiftrole
Description	-	Security configuration	-
Maximum table threshold	-	Database	movie_interactions_glue_db
		Lake Formation configuration	-
		Table prefix	raw_

Below the properties, there is a section for 'Crawler runs (1/1)'. It shows a table with the following columns: Name, Status, and Last updated (UTC). The table contains one entry, which is partially visible.

Once you run crawler it will create a metadata table under movie_interactions_glue_db as below

🔔 **Announcing new optimization features for Apache Iceberg tables**
Optimize storage for Apache Iceberg tables with automatic snapshot retention and orphan file deletion. [Learn more](#) 📄

movie_interactions_glue_db

Last updated (UTC)
September 28, 2025 at 04:48:30  [Edit](#)

Database properties

Name	Description	Location	Created on (UTC)
movie_interactions_glue_db	-	-	September 28, 2025 at 03:5

Tables (2)

Last updated (UTC)
September 28, 2025 at 04:51:47  [Delete](#) [Add tables using crawler](#)

View and manage all available tables.

☐	Filter tables							
☐	Name	Database	Location	Classification	Deprecated	View data	Data quality	Color
☐	raw_events_data_bi	movie_interactions	s3://events-data-bi	CSV	-	Table data	View data quality	View

Edit schema to match redshift schema in table in glue database->table,

us-east-1.console.aws.amazon.com/glue/home?region=us-east-1#/v2/data-catalog/tables/edit-schema-json/

Search [Alt+S]

Tables > raws_events_data_bucket9c1c > Edit schema as JSON

```
7    "Name": "user_id",
8    "Type": "int"
9  },
10 {
11   "Name": "movie_id",
12   "Type": "int"
13 },
14 {
15   "Name": "interaction_type",
16   "Type": "string"
17 },
18 {
19   "Name": "event_timestamp",
20   "Type": "string"
21 },
22 {
23   "Name": "device_type",
24   "Type": "string"
25 },
26 {
27   "Name": "session_id",
28   "Type": "string"
29 },
30 {
31   "Name": "duration_seconds",
32   "Type": "bigint"
33 }
34 ]
```

INT

Create vpc endpoints and edit security groups:

Endpoints (6) Info

Find endpoints by attribute or tag

Name	VPC endpoint ID	Endpoint type	Status	Service
s3-gateway-endpoint	vpce-02c66b702bbbe779a	Gateway	Available	com.amazonaws.s3
redshift-interface-endpoint	vpce-0c74dec2f2e04b9c9	Interface	Available	com.amazonaws.redshift
sts-interface-endpoint	vpce-01cf5e74fc2e7cabe	Interface	Available	com.amazonaws.sts
secretsmanager-interface-endpoint	vpce-071d250cdb0936396	Interface	Available	com.amazonaws.secretsmanager

Select an endpoint

Edit inbound rules Info

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules Info

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-0498c251c7f6abcf	All TCP	TCP	0 - 65535	Custom	

Add rule

Cancel Preview changes

Edit outbound rules Info

Outbound rules control the outgoing traffic that's allowed to leave the instance.

Outbound rules Info

Security group rule ID	Type	Protocol	Port range	Destination	Description - optional
sgr-099e67e1ee940930f	All TCP	TCP	0 - 65535	Custom	
sgr-0cc32454a6d6a4309	HTTPS	TCP	443	Custom	

Add rule

Cancel Preview changes Save

Glue connection to connect to redshift:

10. Navigate back to **Endpoints**, and wait until all endpoints have a **Status of Available**. Click **Refresh** periodically until they do.

Create a Glue connection

1. Navigate to **AWS Glue**, toward the right side, a bit further down, click on **Go to connections**, then click on **Create connection**.
2. For the Data source, select **Amazon Redshift**, and in the bottom right click **Next**.
3. At the **Configuration connection** page, fill in the following values:
 - **Database instances:** Select **redshiftclusterdemo**.
 - **Password:** Paste in the password that you stored earlier.
 - **IAM service role:** Select the **GlueRedshiftRole**.
4. Click **Next**, then click **Next** again.
5. Click **Create connection**.

Use password here which you created in redshift properties

The screenshot shows the 'Configure connection' page in the AWS Glue console. On the left, a navigation pane lists various AWS services. The main area is titled 'Configure connection' and contains a 'Connection details' section. This section includes a dropdown for 'Database instances' (set to 'redshiftclusterdemo-ccnc3dm3d3sy'), a text field for 'Database name' (set to 'interactions_db'), and a 'Credential type' section with radio buttons for 'Username and password' (selected) and 'AWS Secrets Manager'. Below this are text fields for 'Username' (set to 'admin') and 'Password' (masked with dots). There is an unchecked checkbox for 'Enforce SSL - optional'. At the bottom, the 'IAM service role' dropdown is set to 'GlueRedshiftRole'. At the bottom right of the form, there are 'Cancel' and 'Previous' buttons.

Step 1: Choose data source
Step 2: **Configure connection**
Step 3: Set properties
Step 4: Review and create

Configure connection

Connection details

Database instances
Provisioned Amazon Relational Database Service instances.
redshiftclusterdemo-ccnc3dm3d3sy

Database name
interactions_db

Credential type
☒ Username and password
☐ AWS Secrets Manager

Username
admin

Password

☐ Enforce SSL - optional

IAM service role
Specify an IAM role from your account that will be used to access AWS Secret Manager and assign IP if VPC is specified.
GlueRedshiftRole

Cancel Previous

Search[Alt+S]

United States (N. Virginia)

mviet

Last modified on 9/28/2025, 12:25:46 AM

Actions

VisualScriptJob detailsRunsData qualitySchedulesVersion Control

+Data source - Data CatalogAWS Glue Data Catalog

Data previewOutput schema

Data preview (16)InfoREADY

Filter sample dataset

interaction_id	user_id	movie_id	interaction_type	event_timestamp	device_type
1	101	1001	click	2025-02-02 10:00:00	mobile
1	101	1001	click	2025-02-02 10:00:00	mobile
2	102	1002	view	2025-02-03 11:00:00	tablet

End session

Previewing 8 of 8 fields

Data source properties - Data Catalog

NameAWS Glue Data Catalog

DatabaseChoose a database.
movie_interactions_glue_db

Tableraw_events_data_bucket9c1c

Partition predicate - optionalEnter a boolean expression supported by Spark SQL, using only partition columns.
Partition predicate syntax for Spark SQL is year == year[date_sub(current_date, 7)] AND month(date_sub(current_date, 7)) AND day == day[date_sub(current_date, 7)].

data

et2

VisualScriptJob detailsRunsData qualitySchedulesVersion Control

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Filter sample dataset

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Previewing 8 of 8 fields

Data source properties - Data Catalog

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You are changing schema and removing duplicates

The screenshot shows the AWS Glue Studio console interface. The top navigation bar includes tabs for Script, Job details, Runs, Data quality, Schedules, and Version Control. The main workspace displays a workflow with two transforms: 'Transform - Change Schema' and 'Transform - Drop Duplicates', both marked with green checkmarks. A sidebar on the left lists various data sources and targets, including Glue Data Catalog, Amazon S3, Amazon Redshift, and Amazon QL. The right sidebar shows the 'Data target properties - Amazon Redshift' configuration, including fields for Name, Node parents, Redshift access type (set to 'Direct data connection - recommended'), and Redshift connection.

And load data to redshift from glue

The screenshot shows the AWS Glue Studio console interface. The top navigation bar includes tabs for Visual, Script, Job details, Runs, Data quality, Schedules, and Version Control. The main workspace displays a workflow with a single node: 'Data target - Amazon Redshift'. The right sidebar shows the 'Redshift connection' configuration, including fields for Connection name, Schema, and Table. The 'Handling of data and target table' section is also visible, with options for APPEND (insert) to target table, MERGE data into target table, TRUNCATE target table, and DROP and recreate target table. A message at the bottom states: 'Target node not supported. You have selected a data target node which is not supported for data preview. Please select another type of node instead.'

Check run:

et2

Visual Script Job details Runs Data quality Schedules Version Control

Target node not supported
You have selected a data target node which is not supported for data preview. Please select another type of node instead.

Data target properties - Amazon Redshift

Name: Amazon Redshift

Node parents: Choose which nodes will provide inputs for this one. Choose one or more parent node

Drop Duplicates (DropDuplicates - Transform)

Redshift access type: ☒ Direct data connection - recommended ☐ Glue Data Catalog tables

Redshift connection: Choose the AWS Glue connection for Amazon Redshift, or create a new connection

Redshift connection: Redshift connection

Connection Database

In redshift:

us-east-1.console.aws.amazon.com/redshiftv2/home?region=us-east-1#query-editor:

Run Save Schedule Clear

Query results Table details

Query 4185

Completed, started on September 28, 2025 at 01:02:29
ELAPSED TIME: 00 m 02 s

Rows returned (10)

Search rows

interaction_id	user_id	movie_id	type	event_time	device_type	session_id	duration_seconds
1	101	1001	click		mobile	session_001	120
2	102	1002	view		tablet	session_002	300
3	101	1001	click		mobile	session_003	180
4	103	1003	view		desktop	session_004	150
5	104	1004	click		mobile	session_005	220
6	105	1005	view		tablet	session_006	250
7	106	1006	click		desktop	session_007	200
8	107	1007	view		mobile	session_008	310
9	108	1008	click		tablet	session_009	130